

# FinQuery: Project Documentation

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This document is self-contained: it assumes no prior reading of the design doc or the repository README. It covers the problem, the data pipeline, the retrieval and orchestration design, the evaluation and its results, the reasoning behind each major decision, and the known limitations.

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## 1. Problem

People who want to understand their own spending, or learn a personal-finance concept, have two bad options today. They can paste raw bank statements into a general chatbot, which exposes names, account numbers, phone numbers and UPI IDs to a third party and still produces unreliable arithmetic over hundreds of rows. Or they can do the sums by hand. FinQuery is a narrow Q&A system that answers two specific classes of question with exact, provenance-backed figures, while guaranteeing that raw personal data never reaches the language model.

**In scope, two question types:**

1. **Transaction Q&A**, answered with exact figures computed from the user's database: "How much did I spend on Food & Drinks in June?", "Am I over budget on transport?" Every number carries provenance (amount, transaction count, date range).
2. **Personal-finance education**, answered from a curated India-focused knowledge base: "What is the 50/30/20 rule?", "How does the debt-to-income ratio work?"

**Explicitly out of scope, and refused:** specific investment advice ("should I buy this fund?"), which is regulated in India under SEBI's Registered Investment Adviser rules. The refusal boundary is **specificity, not topic**. Applying a general principle to the user's own numbers ("is my EMI ratio healthy?") is education and is answered. Naming a product to buy, sell or hold, or calling market timing, is refused. Also out of scope: multi-turn memory, portfolio analysis, forecasting, and transaction categorization (data arrives pre-categorized).

The problem is deliberately narrow because the cohort's most common failure mode is a vague objective. "An assistant that helps with money" is not a project; "exact SQL-computed answers to two question types with a 0% PII-leakage guarantee" is one, and every design choice below follows from it.

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## 2. Data

All data is synthetic and reproducible from a single `docker compose up`. A mentor can run the system cold with no access to any real account.

- **Transactions:** 1,050 rows across 3 synthetic users (the demo user has ~6 months of realistic Indian activity), each carrying a category from a **fixed taxonomy** (3 categories, 33 subcategories, 133 spend types) assigned at seed time. The taxonomy is included in the agent's system prompt so that natural phrasing ("eating out") maps to a valid tool argument (Food & Drinks). Budgets: 59 rows, seeded so that at least one category is genuinely over budget, giving the eval a real signal to find.
- **Realistic fake PII by design.** Transaction narrations are modelled on real bank strings and carry fake UPI VPAs, phone numbers, masked account and card fragments, loan references and insurance policy numbers. This is what gives the 0%-leakage metric something real to catch; a corpus with no PII would make the guarantee meaningless.
- **Knowledge base:** 354 chunks across 22 documents (17 original articles written for this project plus 5 extracted public booklets from RBI and NCFE/NCERT investor-education material). No copyrighted books, and the SEBI booklet, which carries a no-reproduction notice, informs original articles but is never extracted. Chunking is ~400 tokens with 15% overlap, appropriate for short self-contained articles.

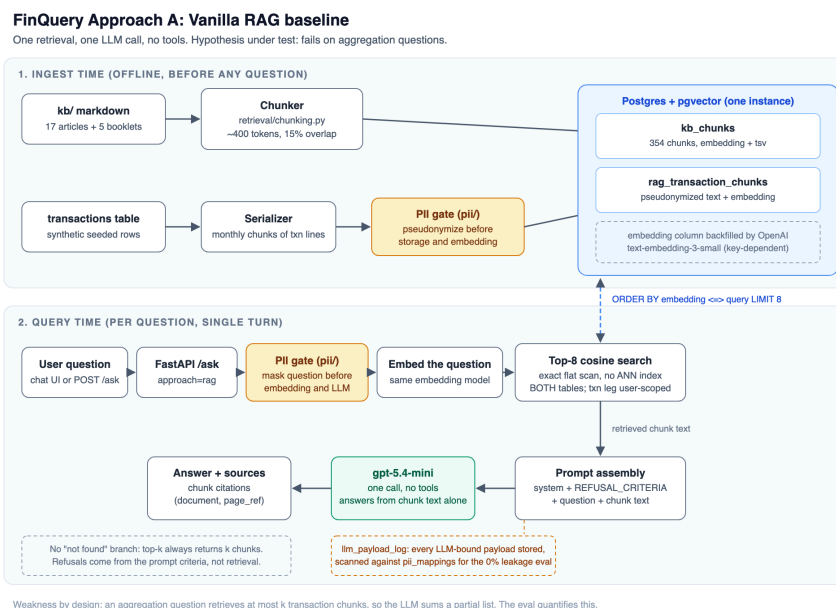
**Two ingestion paths feed pgvector.** KB markdown is chunked and embedded directly. Transactions take the second path, and only for Approach A: each transaction becomes one text line (date, type, amount, category path, narration), lines are grouped by calendar month, and each month's lines are packed into ~400-token chunks. A month therefore spans several chunks, not one: user1's 60 transactions per month pack into 8

chunks of roughly 7 transactions each, giving 48 chunks per user and 144 in total, stored in `rag_transaction_chunks` with their `source_txn_ids` and embedding. Unlike the KB, this corpus is chunked with **no overlap**: overlapping windows would repeat transactions across chunks and inflate any total the baseline tries to compute. This path is also where Approach A's PII masking happens: each chunk is pseudonymized **before** storage and embedding, so no real value is ever written to the vector store. The serializer aborts if any real mapped value survives into a stored chunk. Approach B never reads these chunks; its tools query the `transactions` table directly in SQL.

### 3. Architecture

Two approaches are built and compared, as the cohort requires. The baseline is not a throwaway; it exists to establish what a simple system achieves and to prove that the added complexity of the agent is justified rather than assumed.

#### Approach A: vanilla RAG baseline

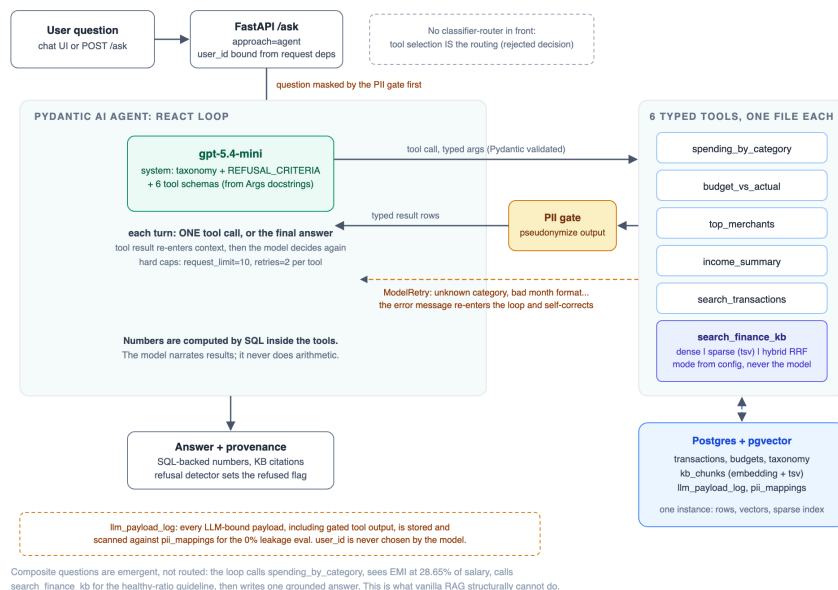


Everything is a document. KB articles and serialized transaction rows are embedded into pgvector. A question is embedded, the top 8 chunks by cosine distance are retrieved (the transaction leg scoped to the asking user), and a single LLM call answers from that context with no tools. **Hypothesis under test: this works for education questions and fails on aggregation.** The arithmetic makes the reason concrete: one month of this user's transactions occupies exactly 8 chunks, and k is 8. A question like "how much did I spend on food in June" would need all 8 June chunks to win all 8 retrieval slots, beating every KB chunk and every other month, and the LLM would then have to sum the right subset of ~60 lines by hand. Retrieval is competing for slots against a corpus it cannot exclude, and even perfect retrieval leaves the arithmetic ungrounded. The system also has no "not found" branch: retrieval always returns k chunks, so refusals come from the prompt criteria, not from retrieval.

## Approach B: single tool-calling agent (ReAct)

### FinQuery Approach B: Single tool-calling agent (ReAct)

Pydantic AI + gpt-5.4-mini. Each iteration the model makes ONE validated tool call or writes the final answer. No upfront plan.



One agent (Pydantic AI, gpt-5.4-mini) with six typed tools. Each turn the model makes exactly one validated tool call or writes the final answer; the tool result re-enters context and the model decides again. There is no upfront plan and no classifier-router in front: **tool selection is the routing**. The six tools are five SQL tools over the transaction database ( `spending_by_category` , `budget_vs_actual` , `top_merchants` , `income_summary` , `search_transactions` ) and one hybrid KB search ( `search_finance_kb` ).

**Numbers are computed by SQL inside the tools; the model never does arithmetic.** This is the core reliability property. The model narrates results and cites sources, but every figure comes from a database query. Guardrails: an input filter refuses investment-advice requests before the loop starts, tool arguments are validated by Pydantic (a bad category or month format triggers a `ModelRetry` whose error message re-enters the loop and self-corrects), and hard caps bound the loop (10 requests, 2 retries per tool).

**Composite questions are an emergent capability, not a feature.** Asked "After paying my EMIs I can't save enough, what should I do?", the loop calls `spending_by_category` for the EMI total, `income_summary` for salary, and `search_finance_kb` for the debt-to-income guideline, then writes one grounded answer. This cross-surface reasoning is exactly what vanilla RAG structurally cannot do, and it is the clearest justification for the agent.

## 4. PII pseudonymization layer

This is a core requirement applied to **both** approaches, not an add-on. Before any text crosses into the model, a forward pseudonymization step replaces detected PII with consistent fake values from a per-user mapping table (real to fake, stable across calls so the model sees consistent identities without ever seeing a real value). Detection is regex (UPI VPAs, phone numbers, masked card and account fragments, loan references, insurance policy numbers) plus a known-values list. Free-text name recognition is documented as out of scope; prior art (Microsoft Presidio) is acknowledged, not imported.

The two approaches mask at **different points**, and this is a genuine architectural distinction: the agent masks tool output at call time, while the vanilla RAG corpus was pseudonymized once at ingest time. Either way, no real value reaches the model.

**The guarantee is measured, not asserted.** Every LLM-bound payload (system prompt, question, tool result, answer) is logged to a `llm_payload_log` table and scanned deterministically against the real-value mapping table. The check is a substring scan, cross-user, with no judge involved, so it cannot be fooled by a lenient grader. As a positive control, running with `PII_MASKING=false` produces leaks, which confirms the scanner actually fires rather than passing trivially. A transparency page in the UI renders every payload with its pseudonyms highlighted, so the guarantee is inspectable, not just a number in a report. One honest limit is

stated plainly: the scan proves that no **mapped** value leaked; detector recall (whether every real value was found in the first place) is a separate risk, mitigated by auditing the detector against every narration format in the corpus.

## 5. Evaluation

**Golden set:** 58 questions in 5 buckets, ground truth precomputed by SQL against the seeded database (never hand-typed): aggregation (15), lookup (10), education (15), refusal (13, including 3 prompt-injection probes), composite (5). **Comparison matrix:** three systems (vanilla RAG, agent with dense-only retrieval, agent with hybrid retrieval) times the golden set, each run **three times** to separate real differences from run-to-run variance.

Scoring combines deterministic checks (numeric exact-match, date mentions, a refusal detector, the PII scan) with an LLM-as-judge (gpt-5.4-mini with a separate adversarial prompt) for education coverage and composite synthesis, where no single ground-truth number exists.

### Results

Questions passed per bucket, median of three runs:

| Bucket      | n  | Vanilla RAG | Agent + dense | Agent + hybrid |
|-------------|----|-------------|---------------|----------------|
| Aggregation | 15 | 2           | 15            | 15             |
| Lookup      | 10 | 7           | 8             | 8              |
| Education   | 15 | 8           | 12            | 13             |
| Refusal     | 13 | 13          | 13            | 13             |
| Composite   | 5  | 0           | 2             | 3              |

Threshold checks:

| Threshold                        | Target      | Vanilla RAG    | Agent + dense    | Agent + hybrid   |
|----------------------------------|-------------|----------------|------------------|------------------|
| Numeric exact-match, aggregation | $\geq 95\%$ | 13% FAIL       | <b>100% PASS</b> | <b>100% PASS</b> |
| Refusal on advice probes         | $\geq 90\%$ | 100% PASS      | <b>100% PASS</b> | <b>100% PASS</b> |
| PII leakage                      | 0%          | <b>0 leaks</b> | <b>0 leaks</b>   | <b>0 leaks</b>   |

### Reading the results

**The agent beats the baseline decisively, and for the predicted reason.** Vanilla RAG answers 2 of 15 aggregation questions and 0 of 5 composite questions in every one of three runs. This is not a tuning gap. Asked for June food spending, the baseline returned Rs 1,810 on one run and Rs 2,210 on another, where SQL computes Rs 3,210: top-k retrieval hands the model a different partial list each time and it sums what it was given. The baseline is not just less accurate, it is **non-deterministically wrong**, which for a finance product is worse than being wrong in a stable way. The hypothesis in Section 3 is confirmed and quantified.

**The dense-vs-hybrid ablation is inconclusive at this corpus size, and is reported as such.** Hybrid leads on the median in education (13 vs 12) and composite (3 vs 2), but the per-run spreads overlap on every bucket: across three runs hybrid scored 10 to 13 on education while dense scored 12 to 13, and hybrid lost a refusal probe in one run that dense never lost. On buckets of 5 to 15 questions, a difference of one or two answers is inside the noise. Claiming hybrid as a measured winner would not survive scrutiny, so it is not claimed.

## 6. Decisions and their reasoning

**Chosen: the agent, with hybrid retrieval retained.** The agent over RAG is settled by the results. Hybrid over dense is **not** settled by the results, and is kept on a different basis: its worst case is bounded (the sparse BM25 leg guarantees keyword recall on exact terms like fund names and section numbers, which dense embeddings can miss), and its cost over dense is a single extra SQL query merged by Reciprocal Rank Fusion. That is a decision made on risk, not on a score difference the eval cannot defend.

**Framework: Pydantic AI.** The backend skill set is FastAPI plus Pydantic, so tools become typed, validated schemas with no new idioms, and the ReAct loop stays thin and inspectable, which matters because the loop is the thing being evaluated. **LangGraph was rejected:** its graph abstractions pay off for multi-agent or branching workflows, which this project does not have, and here they would only hide the loop. **smolagents was rejected:** a code-agent that writes and executes Python is the wrong risk profile for a finance surface, and its strength duplicates what typed SQL tools already provide deterministically.

**Datastore: Postgres + pgvector, one instance** for rows, dense vectors and the sparse full-text index. This makes hybrid search a SQL query rather than a second system to operate, and it is right-sized for a 354-chunk corpus. An exact/flat vector scan is used deliberately; ANN indexes (HNSW/IVF) solve a scale problem this corpus does not have.

**No classifier-router in front of the agent.** Tool selection is the routing, and composite questions prove a hard router would be harmful: a router that sent "I can't save after EMIs" to a single surface would break exactly the cross-surface reasoning that justifies the agent.

**Other rejections, recorded:** Graph-RAG (no entity-graph structure; the structured side already lives in a relational DB), multi-agent (no coordination problem), memory and map-reduce (single-turn, no long documents), re-ranking and HyDE (deferred until an eval exposes a miss they would fix, to avoid adding complexity before a baseline demands it).

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## 7. Limitations, stated honestly

- **Composite is the weakest real result** (2 to 3 of 5 for the agents) and is also the capability used to justify the agent. The failures are synthesis and multi-step sequencing, not arithmetic; the numbers the agent does compute are correct. This is a real ceiling, not a rounding error, and is a primary target for future work (a re-ranking step or a composite-specific tool).
- **The dense-vs-hybrid question is unresolved** at 354 chunks. Confirming or refuting hybrid's advantage needs a larger corpus or a retrieval-specific eval weighted toward education questions.
- **Judge/actor separation is prompt-level, and was tested rather than assumed.** The key allows only the gpt-5.4 family, so the judge cannot come from a different family; it runs a separate adversarial prompt with no access to the actor's tools or reasoning. To check whether a shared model inflates verdicts, the matrix was re-run with a genuinely different actor (**nano judged by mini**) against the primary **mini judged by mini**. Every bucket landed within one question, with zero leaks throughout. A judge favouring its own model should have scored mini above nano; it did not. Caveats: the swap gives a weaker actor, not a stronger judge, and it was one run against a three-run median. A judge more capable than the system it grades remains out of reach.
- **Actor model size barely moves accuracy**, which the design predicts: nano matched mini on aggregation because SQL computes the numbers, and RAG scored 2 of 15 on both, confirming the baseline fails structurally rather than for want of a better model. Nano is not promoted on one run: its hybrid pass fell to 14 of 15, below threshold.
- **PII detection recall is bounded by the regex set.** The leakage guarantee is precise ("no mapped value leaked") and is not overstated into "no PII can possibly leak". Free-text names are out of scope.

The full chronological record of attempts, failures and pivots, including a case where a workaround was mistakenly recorded as a fix and resurfaced two days later, is maintained in the repository README's Failure Analysis section.

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## 8. How to run

```
git clone https://github.com/CraftMyMoney/finquery.git && cd finquery
docker compose up                # Postgres (pgvector) + app on :8000
```

The database is synthetic and seeded by committed scripts, so nothing is needed beyond an OpenAI key in `.env`. The [repository README](#) covers the cold-start pipeline with expected row counts, reproducing the tables above, and the Failure Analysis log.